

Sad or hopeless & teen smoking

Group SadOrHopeLess * Response CurrentCigaretteUse

THE METHOD

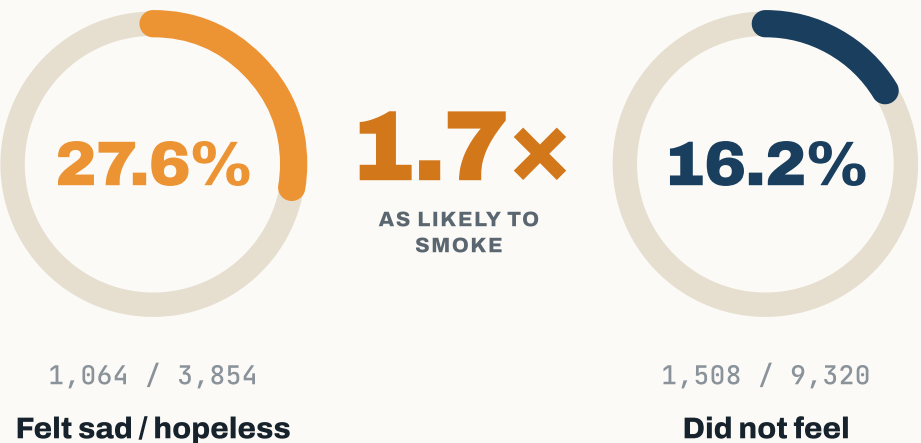
Two-proportion z-test

$H_0: p_{\text{yes}} - p_{\text{no}} = 0$ · $H_1: p_{\text{yes}} - p_{\text{no}} \neq 0$
 $\alpha = 0.05$ · two-sided · $\text{diff} = p_{\text{yes}} - p_{\text{no}}$

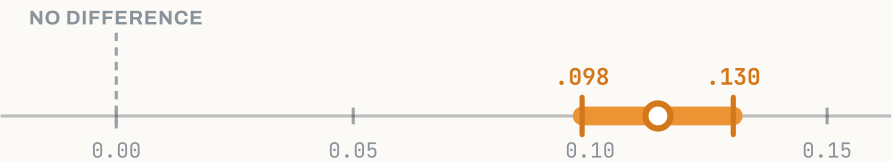
MAIN RESEARCH QUESTION · Two-proportion z-test

Is the proportion of current cigarette use different between students who felt sad or hopeless and those who did not?

01 Current cigarette use by group



02 Difference in proportions 95% confidence interval ($p_{\text{yes}} - p_{\text{no}}$)



diff +0.114

z 15.05

p < .0001

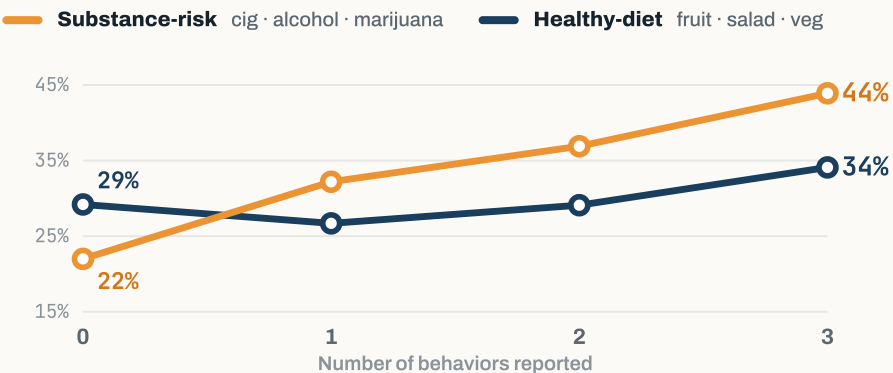
$\alpha = .05$ Reject H_0

Current smoking is **significantly higher** with distress — a +11.4 pp gap, with the 95% CI entirely above zero.

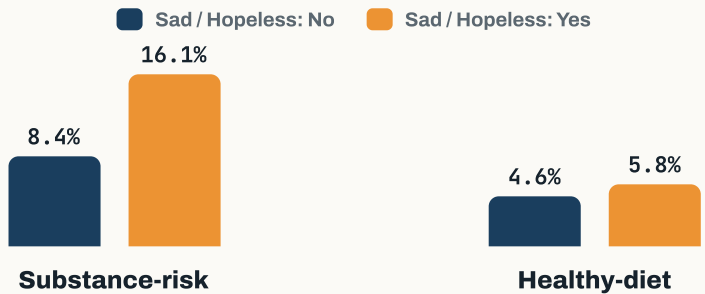
ADDITIONAL EDA QUESTION · Exploratory

Do students who felt sad or hopeless show different clustering patterns between substance-related risk behaviors and healthy diet behaviors?

01 Sad / hopeless % as behavior count rises (0→3)



02 Reporting all three behaviors at once



Risk behaviors **nearly double** with distress — healthy habits stay flat.